

# ADVANCED

## Stage 6

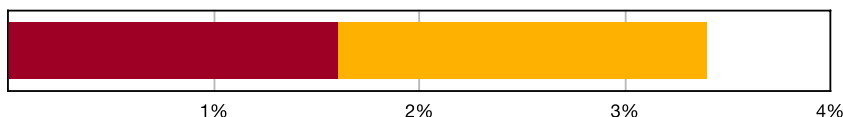
### Calculus (Adv), C1 Introduction to Differentiation (Adv)

#### Standard Differentiation (Y11)

**Teacher:** SHS Maths

**Exam Equivalent Time:** 51 minutes (based on allocation of 1.5 minutes per mark)

## C1 Introduction to Differentiation C2 Differential Calculus



**\*SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

### HISTORICAL CONTRIBUTION

- C1 Introduction to Differentiation* is Y11 content that has contributed an average of 1.6% per exam since the new syllabus was introduced in 2020.
- This topic has been split into three sub-topics for analysis purposes: 1-*Standard Differentiation* (0.8%), 2-*Tangents* (0.8%), and 3-*Rates of Change*.
- This analysis looks at the sub-topic *Standard Differentiation*.

### HSC ANALYSIS - What to expect and common pitfalls

- Standard Differentiation* (0.8%) is the critical gateway content for all the calculus to follow in the Advanced course. It was last examined in a dedicated multiple choice question in both 2023 and 2022.
- This topic area looks at the product, quotient and chain rules, but importantly excludes log, exponential and trig underlyings which are covered in later topics.
- Standard Differentiation* typically provides low hanging fruit for students, although 2023 Adv 7 MC was a very challenging chain rule/composite function example that deserves specific attention.
- First principles and limit equations are covered in the database and should be reviewed.
- TIP: Minimise errors by creating a clear cut process for writing formulae and identifying inputs before any calculations are made using the quotient rule.

### Questions

#### 1. Calculus, 2ADV C1 EQ-Bank 1 MC

The derivative of  $nx^{2n+1}$  can be expressed as

- $2n^2x^{2n+1}$
- $2n^2x^{2n}$
- $(2n+1)nx^{2n}$
- $(2n+1)nx^{2n+1}$

#### 2. Calculus, 2ADV C1 2022 HSC 5 MC

Let  $h(x) = \frac{f(x)}{g(x)}$ , where

$$\begin{aligned} f(1) &= 2 & f'(1) &= 4 \\ g(1) &= 8 & g'(1) &= 12 \end{aligned}$$

What is the gradient of the tangent to the graph of  $y = h(x)$  at  $x = 1$ ?

- 8
- 8
- $-\frac{1}{8}$
- $\frac{1}{8}$

#### 3. Calculus, 2ADV C1 2023 HSC 7 MC

It is given that  $y = f(g(x))$ , where  $f(1) = 3$ ,  $f'(1) = -4$ ,  $g(5) = 1$  and  $g'(5) = 2$ .

What is the value of  $y'$  at  $x = 5$ ?

- 8
- 4
- 3
- 6

#### 4. Calculus, 2ADV C1 2023 MET2 11 MC

Two functions,  $f$  and  $g$ , are continuous and differentiable for all  $x \in \mathbb{R}$ . It is given that  $f(-2) = -7$ ,  $g(-2) = 8$  and  $f'(-2) = 3$ ,  $g'(-2) = 2$ .

The gradient of the graph  $y = f(x) \times g(x)$  at the point where  $x = -2$  is

- 6
- 0
- 6
- 10

5. Calculus, 2ADV C1 2016 HSC 11b

Differentiate  $\frac{x+2}{3x-4}$ . *(2 marks)*

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6. Calculus, 2ADV C1 EQ-Bank 2

When differentiating  $f(x) = 3 - 2x - x^2$  from first principles, a student began the solution as shown below.

Complete the solution. *(2 marks)*

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

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7. Calculus, 2ADV C1 EQ-Bank 3

- i. Use differentiation by first principles to find  $y'$ , given  $y = 2x^2 + 5x$ . *(2 marks)*
  - ii. Find the equation of the tangent to the curve when  $x = 1$ . *(1 mark)*
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8. Calculus, 2ADV C1 2004 HSC 1b

Differentiate  $x^4 + 5x^{-1}$  with respect to  $x$ . *(2 marks)*

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9. Calculus, 2ADV C1 2005 HSC 2bii

Differentiate with respect to  $x$ :

$$\frac{x^2}{x-1}$$
. *(2 marks)*

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10. Calculus, 2ADV C1 2014 HSC 11c

Differentiate  $\frac{x^3}{x+1}$ . *(2 marks)*

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11. Calculus, 2ADV C1 2015 HSC 12c

Find  $f'(x)$ , where  $f(x) = \frac{x^2+3}{x-1}$ . *(2 marks)*

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12. Calculus, 2ADV C1 2019 HSC 11c

Differentiate  $\frac{2x+1}{x+5}$ . *(2 marks)*

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13. Calculus, 2ADV C1 2013 HSC 11b

Evaluate  $\lim_{x \rightarrow 2} \frac{(x-2)(x+2)^2}{x^2-4}$ . *(2 marks)*

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14. Calculus, 2ADV C1 EQ-Bank 6

Use the definition of the derivative,  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$  to find  $f'(x)$  if  $f(x) = 5x^2 - 2x$ . *(2 marks)*

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15. Calculus, 2ADV C1 EQ-Bank 7

Differentiate  $x(1+3x)^5$  with respect to  $x$ . *(2 marks)*

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16. Calculus, 2ADV C1 SM-Bank 14

Evaluate  $f'(4)$ , where  $f(x) = x\sqrt{2x+1}$ . *(3 marks)*

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17. Calculus, 2ADV C1 SM-Bank 25

Let  $g(x) = (2-x^3)^3$ .

Evaluate  $g'(-1)$ . *(2 marks)*

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18. Calculus, 2ADV C3 2010 HSC 8d

Let  $f(x) = x^3 - 3x^2 + kx + 8$ , where  $k$  is a constant.

Find the values of  $k$  for which  $f(x)$  is an increasing function. *(2 marks)*

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## Worked Solutions

### 1. Calculus, 2ADV C1 EQ-Bank 1 MC

$$y = nx^{2n+1}$$

$$\begin{aligned}y' &= (2n+1)nx^{2n+1-1} \\ &= (2n+1)nx^{2n}\end{aligned}$$

$$\Rightarrow C$$

### 2. Calculus, 2ADV C1 2022 HSC 5 MC

Using the quotient rule:

$$\begin{aligned}h'(1) &= \frac{g(1) f'(1) - f(1) g'(1)}{[g(1)]^2} \\ &= \frac{8 \times 4 - 2 \times 12}{8^2} \\ &= \frac{32 - 24}{64} \\ &= \frac{1}{8}\end{aligned}$$

$$\Rightarrow D$$

### 3. Calculus, 2ADV C1 2023 HSC 7 MC

$$y = f(g(x))$$

♦♦ Mean mark 38%.

$$\begin{aligned}y' &= f'(g(5)) \times g'(5) \\ &= f'(1) \times 2 \\ &= -4 \times 2 \\ &= -8\end{aligned}$$

$$\Rightarrow A$$

## Worked Solutions

### 4. Calculus, 2ADV C1 2023 MET2 11 MC

Using the Product Rule when  $x = -2$ :

$$\begin{aligned}\frac{d}{dx}(f(x) \times g(x)) &= f(x)g'(x) + g(x)f'(x) \\ &= f(-2)g'(-2) + g(-2)f'(-2) \\ &= -7 \times 2 + 8 \times 3 \\ &= 10\end{aligned}$$

$$\Rightarrow D$$

♦♦♦ Mean mark 22%.

### 5. Calculus, 2ADV C1 2016 HSC 11b

$$y = \frac{x+2}{3x-4}$$

Using the quotient rule:

$$\begin{aligned}\left(\frac{g}{h}\right)' &= \frac{gh' - gh'}{h^2} \\ y' &= \frac{1(3x-4) - (x+2) \cdot 3}{(3x-4)^2} \\ &= \frac{-10}{(3x-4)^2}\end{aligned}$$

### 6. Calculus, 2ADV C1 EQ-Bank 2

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3 - 2(x+h) - (x+h)^2 - (3 - 2x - x^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3 - 2x - 2h - x^2 - 2hx - h^2 - 3 + 2x + x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2h - 2hx - h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(-2x - 2 - h)}{h}\end{aligned}$$

$$\therefore f'(x) = -2x - 2$$

## 7. Calculus, 2ADV C1 EQ-Bank 3

i.  $f(x) = 2x^2 + 5x$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2(x+h)^2 + 5(x+h)) - (2x^2 + 5x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 + 5x + 5h - 2x^2 - 5x}{h} \\ &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + 5h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(4x + 5 + 2h)}{h} \end{aligned}$$

$$\therefore y' = 4x + 5$$

ii. When  $x = 1, y = 7$

$$y' = 4 + 5 = 9$$

$$\therefore y - 7 = 9(x - 1)$$

$$y = 9x - 2$$

## 8. Calculus, 2ADV C1 2004 HSC 1b

$$y = x^4 + 5x^{-1}$$

$$\frac{dy}{dx} = 4x^3 - 5x^{-2}$$

## 9. Calculus, 2ADV C1 2005 HSC 2bii

$$y = \frac{x^2}{x-1}$$

$$\text{Using } \frac{dy}{dx} = \frac{u'v - uv'}{v^2}$$

$$u = x^2 \quad v = x - 1$$

$$u' = 2x \quad v' = 1$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{2x(x-1) - x^2(1)}{(x-1)^2} \\ &= \frac{2x^2 - 2x - x^2}{(x-1)^2} \\ &= \frac{x^2 - 2x}{(x-1)^2} \\ &= \frac{x(x-2)}{(x-1)^2} \end{aligned}$$

## 10. Calculus, 2ADV C1 2014 HSC 11c

$$y = \frac{x^3}{x+1}$$

$$\text{Using } \frac{dy}{dx} = \frac{u'v - uv'}{v^2}$$

$$u = x^3 \quad v = (x+1)$$

$$u' = 3x^2 \quad v' = 1$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{3x^2(x+1) - x^3(1)}{(x+1)^2} \\ &= \frac{3x^3 + 3x^2 - x^3}{(x+1)^2} \\ &= \frac{2x^3 + 3x^2}{(x+1)^2} \\ &= \frac{x^2(2x+3)}{(x+1)^2} \end{aligned}$$

### 11. Calculus, 2ADV C1 2015 HSC 12c

$$f(x) = \frac{x^2 + 3}{x - 1}$$

Using the quotient rule:

$$u = x^2 + 3 \quad v = x - 1$$

$$u' = 2x \quad v' = 1$$

$$\begin{aligned} f'(x) &= \frac{u'v - uv'}{v^2} \\ &= \frac{2x(x - 1) - (x^2 + 3) \times 1}{(x - 1)^2} \\ &= \frac{2x^2 - 2x - x^2 - 3}{(x - 1)^2} \\ &= \frac{x^2 - 2x - 3}{(x - 1)^2} \\ &= \frac{(x - 3)(x + 1)}{(x - 1)^2} \end{aligned}$$

### 12. Calculus, 2ADV C1 2019 HSC 11c

Using quotient rule:

$$u = 2x + 1, \quad v = x + 5$$

$$u' = 2, \quad v' = 1$$

$$\begin{aligned} y' &= \frac{u'v - v'u}{v^2} \\ &= \frac{2(x + 5) - (2x + 1)}{(x + 5)^2} \\ &= \frac{2x + 10 - 2x - 1}{(x + 5)^2} \\ &= \frac{9}{(x + 5)^2} \end{aligned}$$

### 13. Calculus, 2ADV C1 2013 HSC 11b

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)^2}{x^2 - 4} \\ &= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)^2}{(x - 2)(x + 2)} \\ &= \lim_{x \rightarrow 2} (x + 2) \\ &= 4 \end{aligned}$$

**COMMENT:** This question has been simplified as students no longer need to factorise the difference between 2 cubes ( $x^3 - 2^3$ ).

### 14. Calculus, 2ADV C1 EQ-Bank 6

$$f(x) = 5x^2 - 2x$$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{5(x + h)^2 - 2(x + h) - (5x^2 - 2x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{5x^2 + 10xh + h^2 - 2x - 2h - 5x^2 + 2x}{h} \\ &= \lim_{h \rightarrow 0} \frac{10xh + h^2 - 2h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(10x + h - 2)}{h} \\ &= \lim_{h \rightarrow 0} 10x + h - 2 \\ &= 10x - 2 \end{aligned}$$

### 15. Calculus, 2ADV C1 EQ-Bank 7

$$y = x(1 + 3x)^5$$

Using the product and chain rules:

$$\begin{aligned} y' &= 1 \times (1 + 3x)^5 + 15x(1 + 3x)^4 \\ &= (1 + 3x)^4(1 + 3x + 15x) \\ &= (1 + 3x)^4(18x + 1) \end{aligned}$$

### 16. Calculus, 2ADV C1 SM-Bank 14

$$f(x) = x\sqrt{2x + 1}$$

$$\begin{aligned} f'(x) &= 1\sqrt{2x + 1} + x \times \frac{1}{2} \times 2(2x + 1)^{-\frac{1}{2}} \\ &= \sqrt{2x + 1} + x(2x + 1)^{-\frac{1}{2}} \\ f'(4) &= \sqrt{9} + 4(9)^{-\frac{1}{2}} \\ &= 3 + \frac{4}{3} \\ &= \frac{13}{3} \end{aligned}$$

17. Calculus, 2ADV C1 SM-Bank 25

Using Chain Rule:

$$\begin{aligned} g'(x) &= 3(2 - x^3)^2(-3x^2) \\ &= -9x^2(2 - x^3)^2 \\ \therefore g'(1) &= -9(-1)^2[2 - (-1)^3]^2 \\ &= -81 \end{aligned}$$

18. Calculus, 2ADV C3 2010 HSC 8d

$$\begin{aligned} f(x) &= x^3 - 3x^2 + kx + 8 \\ f'(x) &= 3x^2 - 6x + k \end{aligned}$$

$f(x)$  is increasing when  $f'(x) > 0$

$$\Rightarrow 3x^2 - 6x + k > 0$$

$f'(x)$  is always positive

$\Rightarrow f'(x)$  is a positive definite.

i.e. when  $a > 0$  and  $\Delta < 0$

$$\begin{aligned} a &= 3 > 0 \\ \Delta &= b^2 - 4ac \\ \therefore (-6)^2 - (4 \times 3 \times k) &< 0 \\ 36 - 12k &< 0 \\ 12k &> 36 \\ k &> 3 \end{aligned}$$

$\therefore f(x)$  is increasing when  $k > 3$ .

♦♦ Mean mark 28%.  
**MARKER'S COMMENT:** The arithmetic required to solve  $36 - 12k < 0$  proved the undoing of many students.